



K.R. MANGALAM UNIVERSITY
THE COMPLETE WORLD OF EDUCATION

LINUX BASH SCRIPTING LAB

(ENCS356)

Lab file Submitted to

K.R. MANGALAM UNIVERSITY

For

Bachelor Of Technology

In

Computer Science Engineering Specialisation

(Cyber Security)

Submitted By:

Payal Rathore (2301410022)

Faculty:

Dr. Rajesh Kumar



Experiment No. 1

Exploring Linux System Commands, Run Levels (Targets), and Help Utilities.

Problem Statement

Efficient use of the Linux operating system requires a strong understanding of basic system commands, system states, and built-in help mechanisms. Unlike graphical operating systems, Linux relies heavily on command-line utilities for system interaction, administration, and automation.

The problem is to explore and practice essential Linux system commands, understand system run levels / targets, and effectively use Linux help utilities such as `man`, `--help`, and *auto-* completion to access command documentation and usage information.

Aim

To study and execute basic Linux system commands, understand Linux run levels (systemd targets), and use Linux help utilities such as `man`, `--help`, and tab auto-completion.

Theory

Linux Command Line Interface (CLI)

The Linux Command Line Interface allows users to interact with the operating system by typing commands. It provides better control, flexibility, and efficiency compared to graphical interfaces, especially for system administration and automation.

System Run Levels / Targets

- Modern Linux systems use **systemd**, where traditional run levels are replaced by **targets**.

Common systemd targets:

- `graphical.target` – Multi-user mode with GUI
- `multi-user.target` – Multi-user mode without GUI
- `rescue.target` – Single-user rescue mode
- `poweroff.target` – Shutdown system
- `reboot.target` – Reboot system

Linux Help Utilities

Linux provides built-in help tools for learning commands:

- `man` – Displays complete manual pages
- `--help` – Shows brief command usage
- **Tab auto-completion** – Completes commands and file names automatically

Commands Used

Commands	Description
<code>pwd</code>	Display present working directory
<code>ls</code>	List files and directories
<code>cd</code>	Change directory
<code>whoami</code>	Display current user
<code>date</code>	Show system date and time
<code>uptime</code>	Show system running time
<code>man</code>	Display manual pages
<code>--help</code>	Display help for a command
<code>systemctl</code>	Manage system services and targets
<code>runlevel</code>	Display previous and current run level
<code>clear</code>	Clear terminal screen

Procedure / Steps

Step 1: Login:

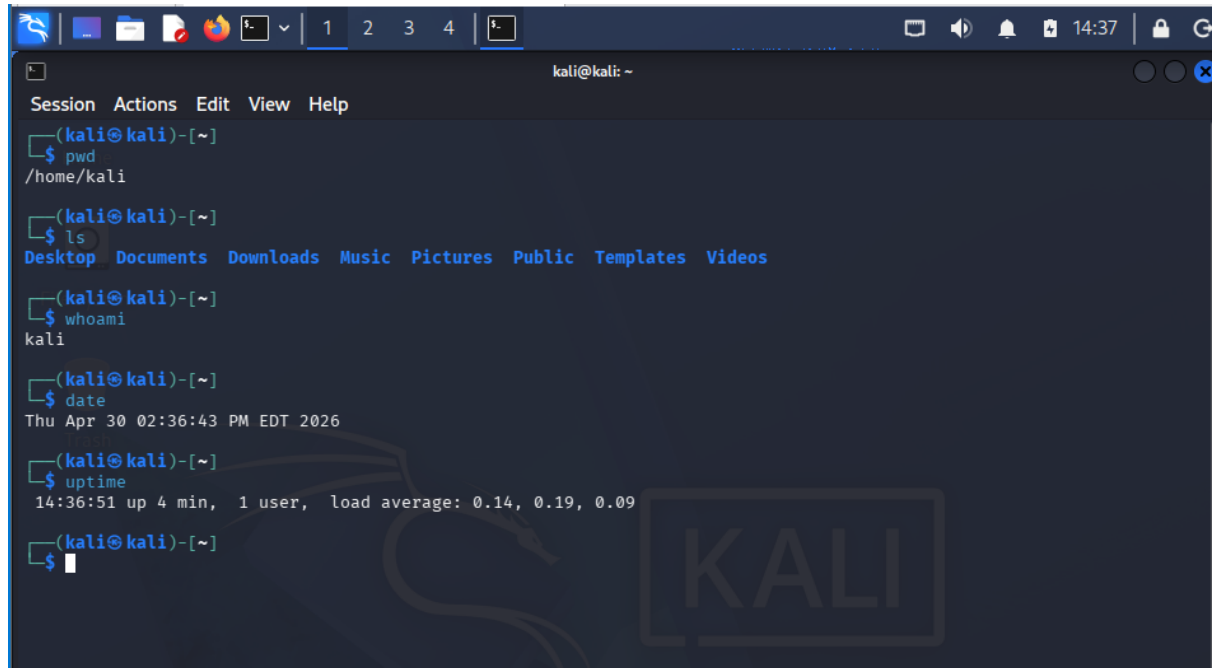
Log in to the Linux virtual machine and open the Terminal.

Step 2: Execute Basic Linux Commands in the Terminal.

Commands:

- `pwd` shows current directory
- `ls` lists files
- `whoami` displays logged-in user
- `date` shows system date/time

- uptime shows how long system is running

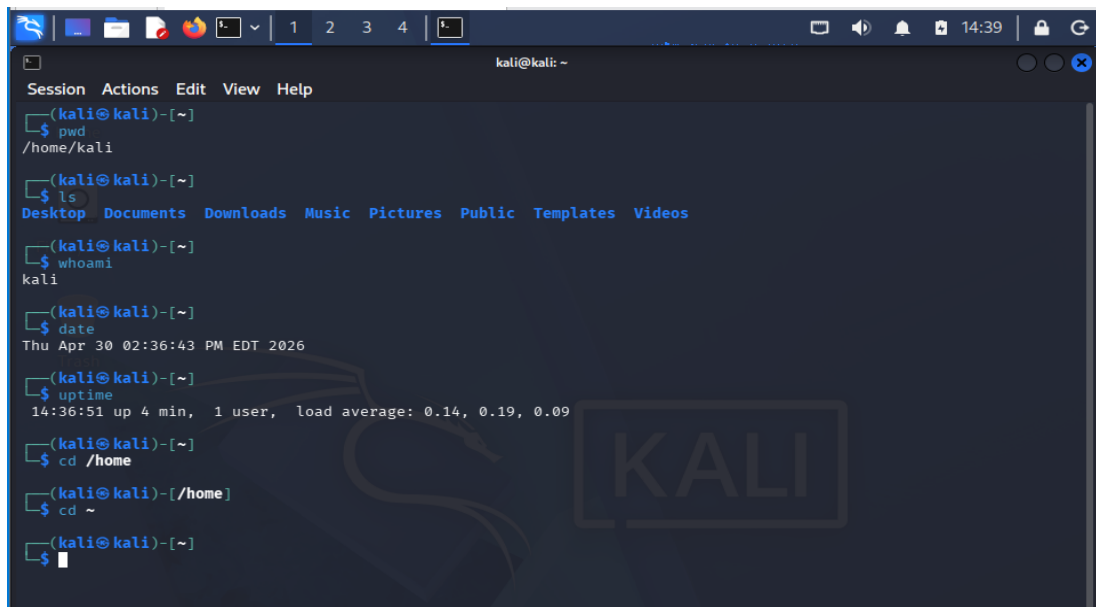


A terminal window titled 'kali@kali: ~' showing the following commands and output:

```
(kali@kali)-[~]  
$ pwd  
/home/kali  
  
(kali@kali)-[~]  
$ ls  
Desktop Documents Downloads Music Pictures Public Templates Videos  
  
(kali@kali)-[~]  
$ whoami  
kali  
  
(kali@kali)-[~]  
$ date  
Thu Apr 30 02:36:43 PM EDT 2026  
  
(kali@kali)-[~]  
$ uptime  
14:36:51 up 4 min, 1 user, load average: 0.14, 0.19, 0.09  
  
(kali@kali)-[~]  
$
```

Step 3: Directory Navigation using:

- `cd /home` moves to home directory.
- `cd ~` returns to user's home directory.



A terminal window titled 'kali@kali: ~' showing the following commands and output:

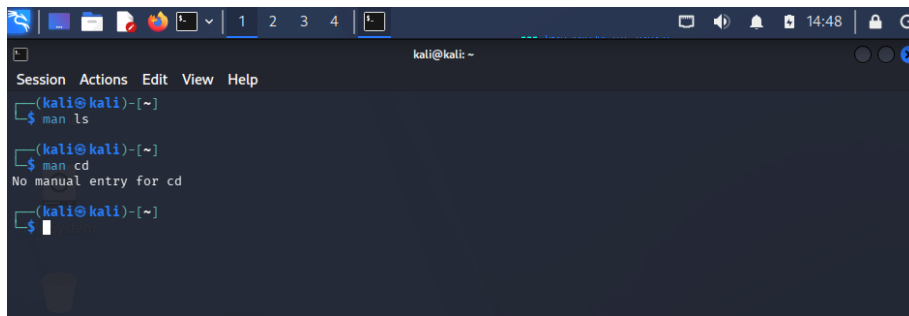
```
(kali@kali)-[~]  
$ pwd  
/home/kali  
  
(kali@kali)-[~]  
$ ls  
Desktop Documents Downloads Music Pictures Public Templates Videos  
  
(kali@kali)-[~]  
$ whoami  
kali  
  
(kali@kali)-[~]  
$ date  
Thu Apr 30 02:36:43 PM EDT 2026  
  
(kali@kali)-[~]  
$ uptime  
14:36:51 up 4 min, 1 user, load average: 0.14, 0.19, 0.09  
  
(kali@kali)-[~]  
$ cd /home  
  
(kali@kali)-[/home]  
$ cd ~  
  
(kali@kali)-[~]  
$
```

Step 4: Using Manual Pages

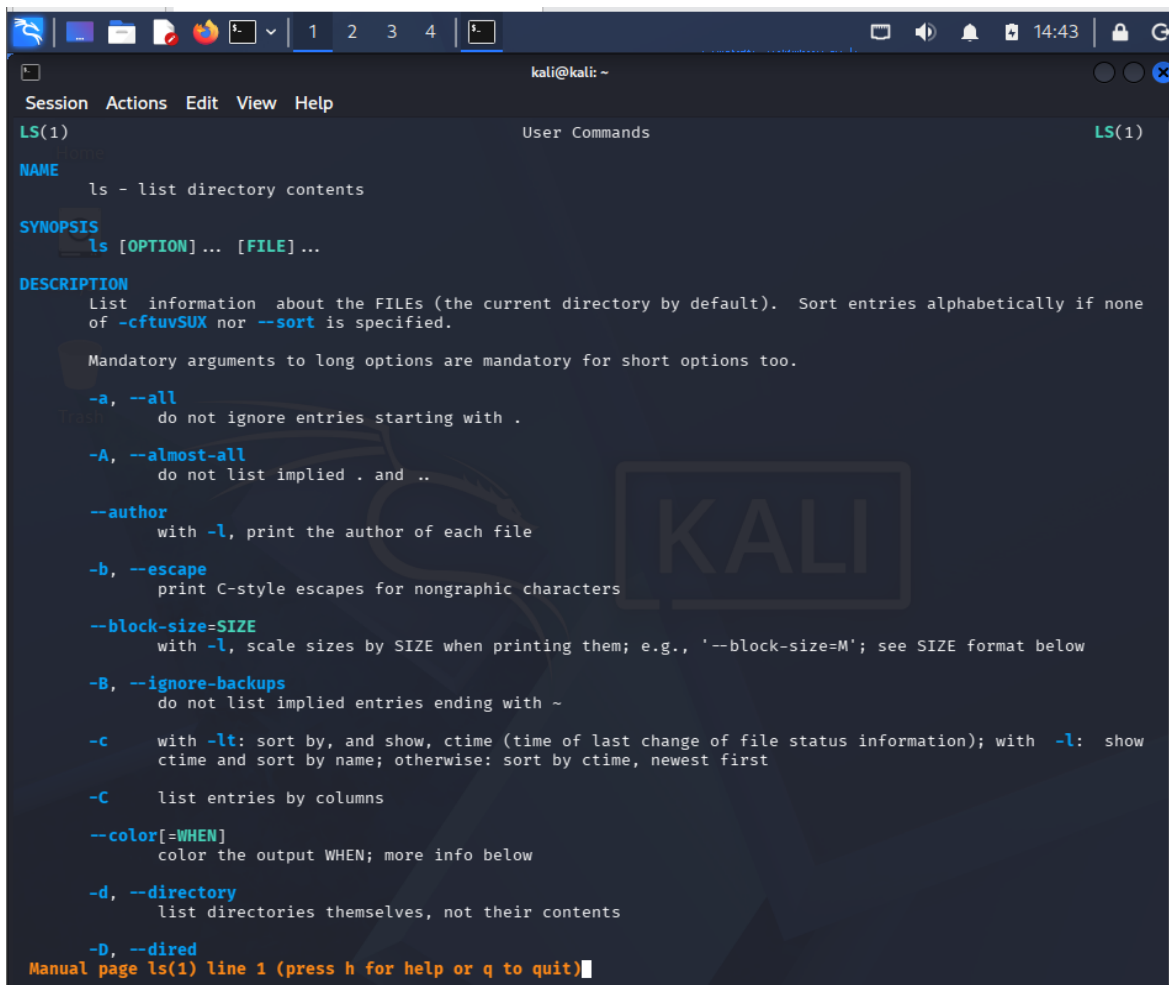
- `man ls`
- `man cd`

It Read command details.

➤ Press `q` to exit the manual.



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ man ls  
(kali@kali)-[~]  
$ man cd  
No manual entry for cd  
(kali@kali)-[~]  
$
```

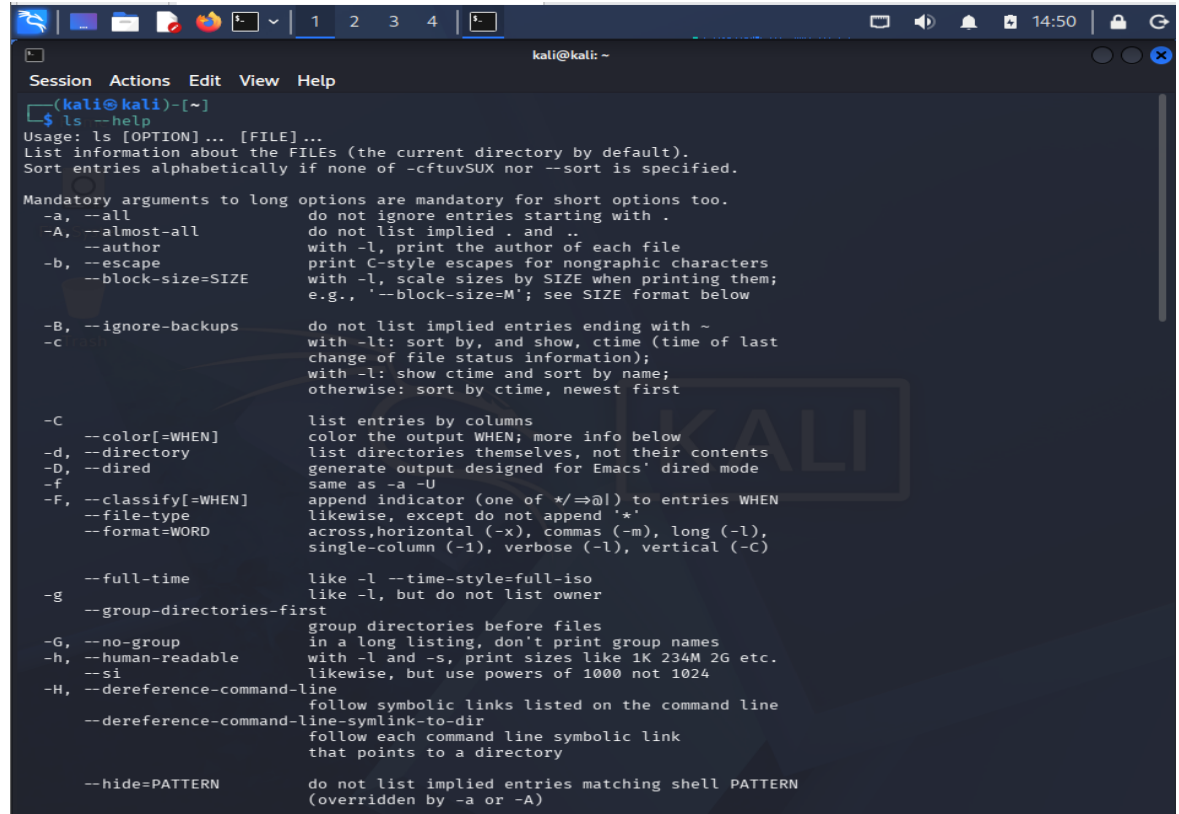


```
kali@kali: ~  
Session Actions Edit View Help  
LS(1) User Commands LS(1)  
NAME  
ls - list directory contents  
SYNOPSIS  
ls [OPTION]... [FILE]...  
DESCRIPTION  
List information about the FILES (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.  
Mandatory arguments to long options are mandatory for short options too.  
-a, --all do not ignore entries starting with .  
-A, --almost-all do not list implied . and ..  
--author with -l, print the author of each file  
-b, --escape print C-style escapes for nongraphic characters  
--block-size=SIZE with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below  
-B, --ignore-backups do not list implied entries ending with ~  
-c with -lt: sort by, and show, ctime (time of last change of file status information); with -l: show ctime and sort by name; otherwise: sort by ctime, newest first  
-C list entries by columns  
--color[=WHEN] color the output WHEN; more info below  
-d, --directory list directories themselves, not their contents  
-D, --dired  
Manual page ls(1) line 1 (press h for help or q to quit)
```

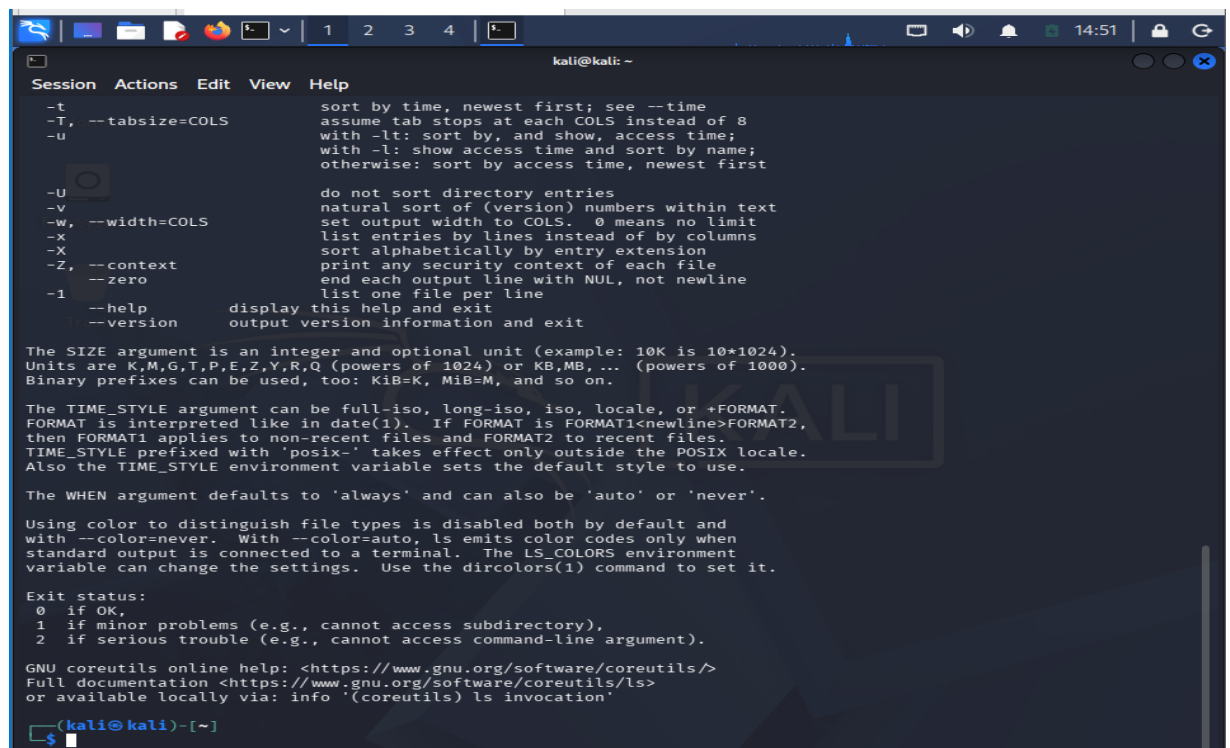
Step 5: Using Help Option:

- `ls -help`
- `systemctl -help`

This displays short usage instructions and optio



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ ls --help  
Usage: ls [OPTION]... [FILE]...  
List information about the FILEs (the current directory by default).  
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.  
  
Mandatory arguments to long options are mandatory for short options too.  
-a, --all                do not ignore entries starting with .  
-A, --almost-all        do not list implied . and ..  
      --author            with -l, print the author of each file  
-b, --escape            print C-style escapes for nongraphic characters  
      --block-size=SIZE  with -l, scale sizes by SIZE when printing them;  
                        e.g., '--block-size=M'; see SIZE format below  
  
-B, --ignore-backups    do not list implied entries ending with ~  
-c                      with -lt: sort by, and show, ctime (time of last  
                        change of file status information);  
                        with -l: show ctime and sort by name;  
                        otherwise: sort by ctime, newest first  
  
-C                      list entries by columns  
      --color[=WHEN]    color the output WHEN; more info below  
-d, --directory        list directories themselves, not their contents  
-D, --dired            generate output designed for Emacs' dired mode  
-f                      same as -a -U  
-F, --classify[=WHEN]  append indicator (one of */=>@|) to entries WHEN  
      --file-type        likewise, except do not append '*'  
      --format=WORD     across, horizontal (-x), commas (-m), long (-l),  
                        single-column (-1), verbose (-l), vertical (-C)  
  
      --full-time        like -l --time-style=full-iso  
      --group-directories-first like -l, but do not list owner  
                        group directories before files  
-G, --no-group          in a long listing, don't print group names  
-h, --human-readable    with -l and -s, print sizes like 1K 234M 2G etc.  
      --si              likewise, but use powers of 1000 not 1024  
-H, --dereference-command-line follow symbolic links listed on the command line  
      --dereference-command-line-symlink-to-dir follow each command line symbolic link  
                        that points to a directory  
  
--hide=PATTERN          do not list implied entries matching shell PATTERN  
                        (overridden by -a or -A)
```



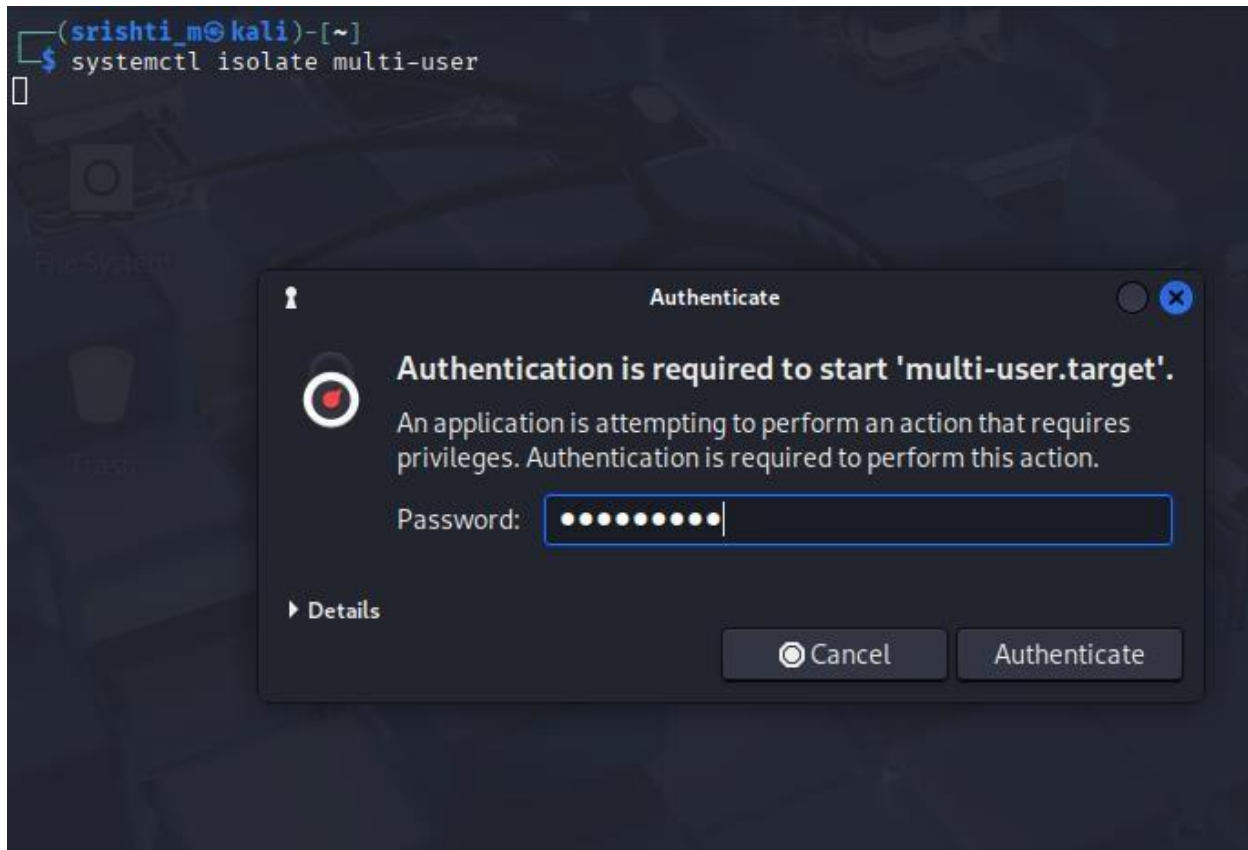
```
kali@kali: ~  
Session Actions Edit View Help  
-t                      sort by time, newest first; see --time  
-T, --tabsize=COLS      assume tab stops at each COLS instead of 8  
-u                      with -lt: sort by, and show, access time;  
                        with -l: show access time and sort by name;  
                        otherwise: sort by access time, newest first  
  
-U                      do not sort directory entries  
-v                      natural sort of (version) numbers within text  
-w, --width=COLS        set output width to COLS. 0 means no limit  
-X                      list entries by lines instead of by columns  
-x                      sort alphabetically by entry extension  
-Z, --context            print any security context of each file  
-z                      end each output line with NUL, not newline  
-1                      list one file per line  
      --help            display this help and exit  
      --version          output version information and exit  
  
The SIZE argument is an integer and optional unit (example: 10K is 10*1024).  
Units are K,M,G,T,P,E,Z,Y,R,Q (powers of 1024) or KB,MB,... (powers of 1000).  
Binary prefixes can be used, too: KiB=K, MiB=M, and so on.  
  
The TIME_STYLE argument can be full-iso, long-iso, iso, locale, or +FORMAT.  
FORMAT is interpreted like in date(1). If FORMAT is FORMAT1<newline>FORMAT2,  
then FORMAT1 applies to non-recent files and FORMAT2 to recent files.  
TIME_STYLE prefixed with 'posix-' takes effect only outside the POSIX locale.  
Also the TIME_STYLE environment variable sets the default style to use.  
  
The WHEN argument defaults to 'always' and can also be 'auto' or 'never'.  
  
Using color to distinguish file types is disabled both by default and  
with --color=never. With --color=auto, ls emits color codes only when  
standard output is connected to a terminal. The LS_COLORS environment  
variable can change the settings. Use the dircolors(1) command to set it.  
  
Exit status:  
0 if OK,  
1 if minor problems (e.g., cannot access subdirectory),  
2 if serious trouble (e.g., cannot access command-line argument).  
  
GNU coreutils online help: <https://www.gnu.org/software/coreutils/>  
Full documentation <https://www.gnu.org/software/coreutils/ls>  
or available locally via: info '(coreutils) ls invocation'  
  
(kali@kali)-[~]  
$
```

Step 6: Check System Run Level / Target

- `runlevel`- “Kali Linux uses systemd, so traditional runlevel commands are deprecated and `systemctl` targets are used instead.”
- `systemctl get-default` shows default system target.

```
(srishti_m@kali)-[~]  
$ runlevel  
unknown  
  
(srishti_m@kali)-[~]  
$ systemctl get-default  
graphical.target
```

```
srishti_m@kali: ~  
File Actions Edit View Help  
(srishti_m@kali)-[~]  
$ systemctl isolate multi-user
```



```
File Actions Edit View Help
status [PATTERN ... |PID ... ]
show [PATTERN ... |JOB ... ]

cat PATTERN ...
help PATTERN ... |PID ...
list-dependencies [UNIT ... ]

start UNIT ...
stop UNIT ...
reload UNIT ...
restart UNIT ...
try-restart UNIT ...
reload-or-restart UNIT ...

Show runtime status of one or more units
Show properties of one or more
units/jobs or the manager
Show files and drop-ins of specified units
Show manual for one or more units
Recursively show units which are required
or wanted by the units or by which those
units are required or wanted
Start (activate) one or more units
Stop (deactivate) one or more units
Reload one or more units
Start or restart one or more units
Restart one or more units if active
Reload one or more units if possible,

(srishti_m@kali)-[~]
$ runlevel
unknown

(srishti_m@kali)-[~]
$ systemctl get-default
graphical.target

(srishti_m@kali)-[~]
$ systemctl isolate graphical.target

(srishti_m@kali)-[~]
$ systemctl isolate graphical.target

(srishti_m@kali)-[~]
$
```

Step 7:- Practice auto-completion by typing partial command names and pressing Tab.

Step 8 :- Clear the terminal screen: Clear

Outcomes

- Successful execution of basic Linux commands
- Display of manual pages and help options
- Identification of current run level/target
- Improved command-line efficiency

Result

Thus, basic Linux system commands, system run levels, and help utilities were successfully explored. The student gained practical understanding of Linux command-line interaction and Linux documentation tools.

Experiment No. 2

Title

File and Directory Management and File Permissions in Linux

Aim

To study and practice Linux commands for file and directory management, understand Linux file and directory permissions, and create and edit files using text editors such as `nano` and `vi`.

Theory

Linux File System

Linux follows a hierarchical file system starting from the root directory `/`. All files and directories are arranged in a tree structure, allowing efficient organization and secure access control.

Files and Directories

- **Files** store data such as text, programs, or scripts.
- **Directories** store files and subdirectories.

File and Directory Permissions

Linux uses permissions to control access:

- **Read (r)** – Read file contents / list directory
- **Write (w)** – Modify file or directory
- **Execute (x)** – Execute file or access directory

Permissions apply to:

- **Owner**
- **Group**
- **Others**

Text Editors

- **nano**: Simple and beginner-friendly text editor
- **vi**: Powerful editor with command and insert modes

Commands Used

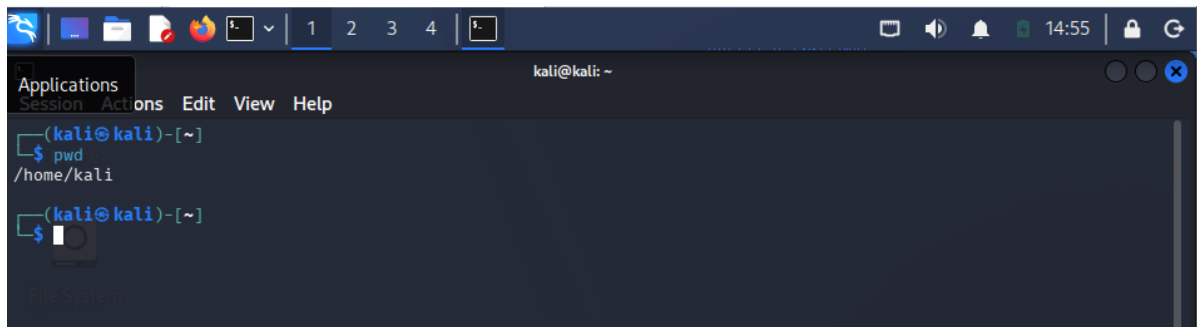
Commands	Description
pwd	Display current directory
ls	List files and directories
ls -l	Display detailed file information
cd	Change directory
mkdir	Create directory
rmdir	Remove empty directory
touch	Create empty file
cat	Display file contents
nano	Edit files using nano editor
vi	Edit files using vi editor
chmod	Change file permissions
stat	Display file status

Procedure / Steps

Step 1: Login

- Log in to the Linux virtual machine and open the **Terminal**.

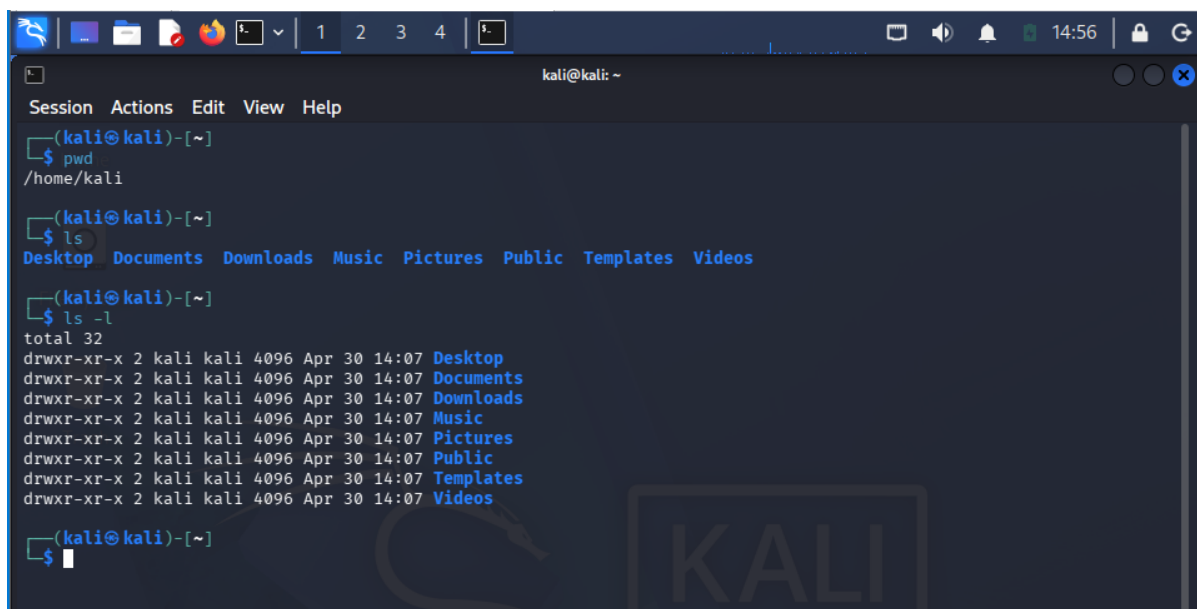
Step 2: Display the Current Directory using `pwd` command.

A terminal window titled 'kali@kali: ~' with a menu bar (Session, Actions, Edit, View, Help). The prompt is '(kali@kali)-[~]'. The command 'pwd' has been entered, and the output is '/home/kali'.

```
(kali@kali)-[~]  
$ pwd  
/home/kali  
  
(kali@kali)-[~]  
$
```

Step 3: List Files and Directories using:

- `ls`
- `ls -l`

A terminal window titled 'kali@kali: ~' with a menu bar (Session, Actions, Edit, View, Help). The prompt is '(kali@kali)-[~]'. The command 'ls' has been entered, and the output shows standard Linux directories. Then, the command 'ls -l' has been entered, and the output shows a detailed listing of the same directories.

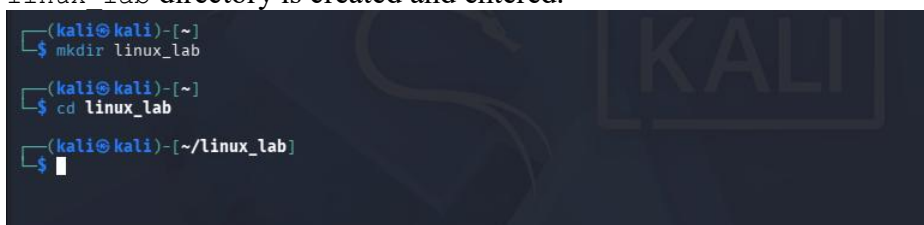
```
(kali@kali)-[~]  
$ pwd  
/home/kali  
  
(kali@kali)-[~]  
$ ls  
Desktop Documents Downloads Music Pictures Public Templates Videos  
  
(kali@kali)-[~]  
$ ls -l  
total 32  
drwxr-xr-x 2 kali kali 4096 Apr 30 14:07 Desktop  
drwxr-xr-x 2 kali kali 4096 Apr 30 14:07 Documents  
drwxr-xr-x 2 kali kali 4096 Apr 30 14:07 Downloads  
drwxr-xr-x 2 kali kali 4096 Apr 30 14:07 Music  
drwxr-xr-x 2 kali kali 4096 Apr 30 14:07 Pictures  
drwxr-xr-x 2 kali kali 4096 Apr 30 14:07 Public  
drwxr-xr-x 2 kali kali 4096 Apr 30 14:07 Templates  
drwxr-xr-x 2 kali kali 4096 Apr 30 14:07 Videos  
  
(kali@kali)-[~]  
$
```

Step 4: Create a New Directory

```
mkdir linux_lab
```

```
cd linux_lab
```

➤ `linux_lab` directory is created and entered.

A terminal window titled 'kali@kali: ~' with a menu bar (Session, Actions, Edit, View, Help). The prompt is '(kali@kali)-[~]'. The command 'mkdir linux_lab' has been entered. Then, the prompt changes to '(kali@kali)-[~/linux_lab]' after the command 'cd linux_lab' is entered.

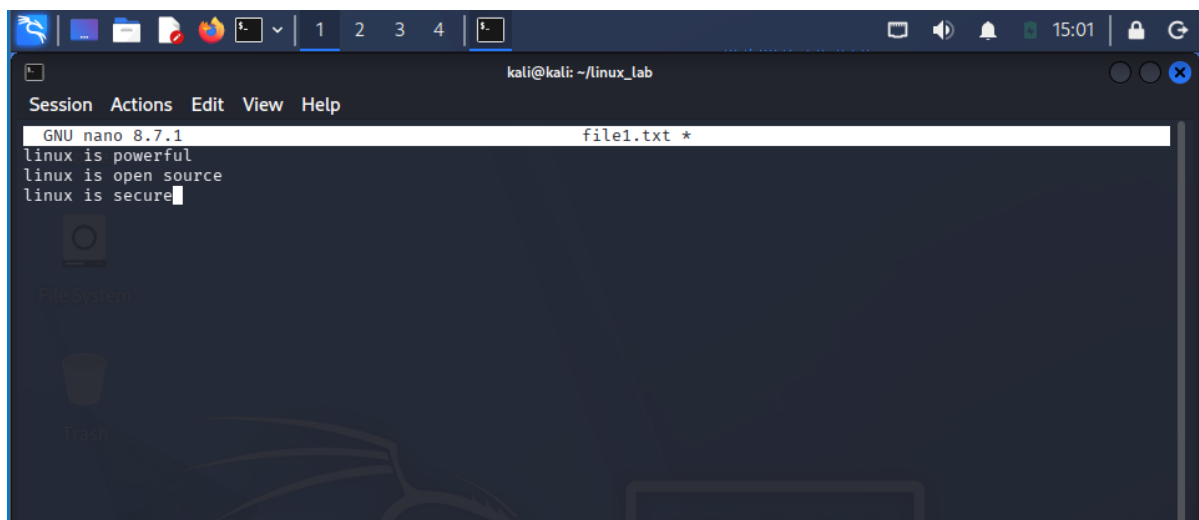
```
(kali@kali)-[~]  
$ mkdir linux_lab  
  
(kali@kali)-[~]  
$ cd linux_lab  
  
(kali@kali)-[~/linux_lab]  
$
```

Step 5: Create Files using : `-touch file1.txt file2.txt`

```
(kali@kali)-[~/linux_lab]
$ touch file1.txt file2.txt
(kali@kali)-[~/linux_lab]
$
```

Step 6: Edit File Using : `nano file1.txt`

- Type some text
- Press **Ctrl + O** → Enter (save)
- Press **Ctrl + X** (exit)



```
(kali㉿kali)-[~/linux_lab]
$ nano file1.txt

(kali㉿kali)-[~/linux_lab]
$
```

Step 7: Edit File Using : vi file2.txt

```
(srishti_m㉿kali)-[~]
$ mkdir linux_lab2

(srishti_m㉿kali)-[~]
$ cd linux_lab2

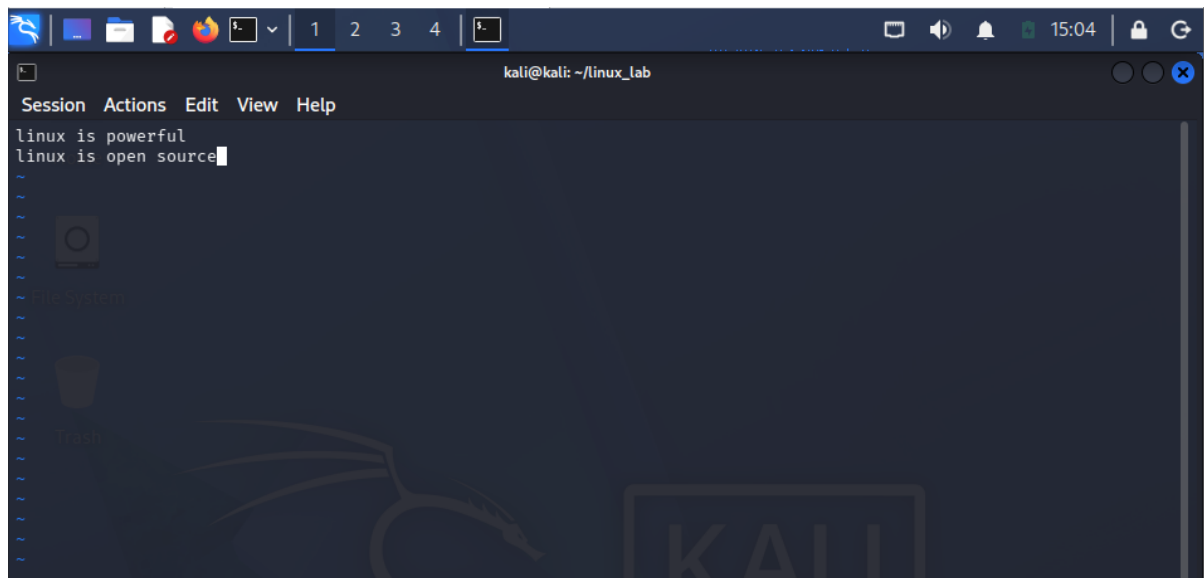
(srishti_m㉿kali)-[~/linux_lab2]
$ touch file1.txt file2.txt

(srishti_m㉿kali)-[~/linux_lab2]
$ nano file1.txt

(srishti_m㉿kali)-[~/linux_lab2]
$ vi file2.txt
```

Then,

- Press **i** to enter insert mode
- Type text
- Press **Esc**
- Type: :wq
- Press Enter



Step 8: Display File Content Using:

➤ `cat file1.txt`



Step 9: Check File Permissions

➤ `ls -l`

```
(srishti_m@kali)-[~/linux_lab2]
$ cat file1.txt

(srishti_m@kali)-[~/linux_lab2]
$ ls -l
total 8
-rw-rw-r-- 1 srishti_m srishti_m 0 Feb 20 02:59 file1.txt
-rw-rw-r-- 1 srishti_m srishti_m 55 Feb 20 03:01 file1.txtn
-rw-rw-r-- 1 srishti_m srishti_m 51 Feb 20 03:07 file2.txt

(srishti_m@kali)-[~/linux_lab2]
$
```

Step 10: Change File Permissions using:

➤ `chmod 755 file1.txt`

```
(srishti_m@kali)-[~/linux_lab2]
$ ls -l
total 8
-rw-rw-r-- 1 srishti_m srishti_m 0 Feb 20 02:59 file1.txt
-rw-rw-r-- 1 srishti_m srishti_m 55 Feb 20 03:01 file1.txtn
-rw-rw-r-- 1 srishti_m srishti_m 51 Feb 20 03:07 file2.txt

(srishti_m@kali)-[~/linux_lab2]
$ chmod 755 file1.txt
```

Step 11: Verify Updated Permissions Using:

- `ls -l file1.txt`
- we will observe permission format: `-rwxr-xr-x`

```
(srishti_m@kali)-[~/linux_lab2]
$ chmod 755 file1.txt

(srishti_m@kali)-[~/linux_lab2]
$ ls -l file1.txt
-rwxr-xr-x 1 srishti_m srishti_m 0 Feb 20 02:59 file1.txt

(srishti_m@kali)-[~/linux_lab2]
$
```

Outcomes

- ✓ Successful creation of directories and files
- ✓ Files edited using `nano` and `vi`
- ✓ File contents displayed using `cat`
- ✓ File permissions viewed and modified using `chmod`

Result

- ❖ Thus, file and directory management commands were successfully executed in Linux. The student learned to create, edit, and manage files and directories, understand file permissions, and use text editors such as `nano` and `vi`.

